

Prove equality of worst-case and asymptotic guarantees in k-unit IID prophet inequalities

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Tightness without Counterexamples: A New Approach and New Results for Prophet Inequalities*.

1 The open problem

This paper addresses an open problem posed by Jiashuo Jiang, Will Ma, Jiawei Zhang in *Tightness without Counterexamples: A New Approach and New Results for Prophet Inequalities* [1] (Economics and Computation (EC), 2023), Page 6, Section 1.2 (New Results), item 2. The website catalogue entry is problem 11 (W4383557893_p0) of the [Open Problems in Operations Research](#) collection.

For every integer $k > 1$, determine whether

$$\lim_{n \rightarrow \infty} \gamma_{k,n} = \inf_{n \geq 1} \gamma_{k,n},$$

where $\gamma_{k,n} := \inf_F \text{ALG}(F, n, k) / \text{Proph}(F, n, k)$ is the tight IID k -unit prophet-inequality ratio for n agents.

Equivalently, decide whether for each $k > 1$ the worst-case IID competitive ratio is achieved (or approached) in the large- n limit, rather than at some finite n .

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_033932_ba740122, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

Overview

This note records partial progress, not a complete solution: the general inequality

$$\gamma_{k,n} \geq L_k \quad (k > 1, n > k)$$

remains unproved. All of the verifier's repairable objections, however, can be fixed, and the precise remaining gap is isolated at the end.

Throughout, F is a nonnegative distribution satisfying

$$0 < P_{n,k}(F) < \infty,$$

which implies $\mathbb{E}X < \infty$. We set $V_{0,k} = P_{0,k} = 0$, and we exclude $F \equiv 0$ as well as undefined ∞/∞ ratios.

2 Existence of L_k

For $1 \leq \ell \leq r$,

$$V_{r,k}(F) \leq \frac{r}{\ell} V_{\ell,k}(F), \quad P_{r,k}(F) \leq \frac{r}{\ell} P_{\ell,k}(F). \quad (1)$$

For the online inequality, run an optimal r -stage policy on an i.i.d. virtual stream, independently mark a uniformly random ℓ -subset of positions as real, and retain only the rewards obtained at real positions. Conditionally on the virtual stream and on all policy decisions, the expected real reward is exactly ℓ/r times the virtual reward, and the construction yields a feasible ℓ -stage policy. The prophet argument is identical after intersecting the top- k set with the random subset.

Now let $m \geq n$, put $Y = BX$ with $B \sim \text{Ber}(n/m)$, and let

$$L \sim \text{Bin}(m, n/m).$$

Revealing the active positions and L to the online policy in advance can only help, so

$$V_{m,k}(Y) \leq \mathbb{E} V_{L,k}(F), \quad P_{m,k}(Y) = \mathbb{E} P_{L,k}(F).$$

By (1), monotonicity in the number of observations, and $V_{0,k} = P_{0,k} = 0$,

$$\mathbb{E} V_{L,k} \leq \left(1 + \frac{\mathbb{E}(L - n)_+}{n}\right) V_{n,k}, \quad \mathbb{E} P_{L,k} \geq \left(1 - \frac{\mathbb{E}(n - L)_+}{n}\right) P_{n,k}.$$

Since $\mathbb{E}L = n$, the two error terms agree. Writing

$$\delta_{m,n} = \frac{\mathbb{E}|L - n|}{2n},$$

we obtain

$$\gamma_{k,m} \leq \frac{1 + \delta_{m,n}}{1 - \delta_{m,n}} \gamma_{k,n}. \quad (2)$$

Moreover,

$$\delta_{m,n} \leq \frac{\sqrt{\text{Var } L}}{2n} \leq \frac{1}{2\sqrt{n}}.$$

Letting $m \rightarrow \infty$, and then $n \rightarrow \infty$ along a subsequence realizing the limit inferior, gives

$$\limsup_m \gamma_{k,m} \leq \liminf_m \gamma_{k,m}.$$

Hence the limit

$$L_k = \lim_{n \rightarrow \infty} \gamma_{k,n} \quad (3)$$

exists.

3 The Poisson comparison, with the missing Bellman details

Let

$$g(s) = \mathbb{E}(X - s)_+;$$

it is decreasing and 1-Lipschitz. If v_j^r denotes the j -choice value with r observations, then

$$v_j^{r+1} = v_j^r + g(v_j^r - v_{j-1}^r). \quad (4)$$

For the marked Poisson process with arrival rate 1, mark law F , and horizon n , let $w_j(t)$ denote its j -choice value. Then

$$\dot{w}_j(t) = g(w_j(t) - w_{j-1}(t)), \quad w_j(0) = w_0(t) = 0. \quad (5)$$

The vector field is globally 2-Lipschitz in the sup norm, so (5) admits a unique global solution.

Put $d_j = w_j - w_{j-1}$, with the formal conventions $d_0 = \infty$ and $g(d_0) = 0$. The cone

$$d_1 \geq d_2 \geq \dots \geq d_k \geq 0 \quad (6)$$

is invariant. Indeed, at a boundary point with $d_j = d_{j+1}$,

$$\frac{d}{dt}(d_j - d_{j+1}) = g(d_j) - g(d_{j-1}) \geq 0,$$

and at $d_j = 0$ we have $\dot{d}_j \geq 0$. Consequently

$$\dot{d}_j = g(d_j) - g(d_{j-1}) \geq 0,$$

so every $d_j(t)$ is nondecreasing and every $g(d_j(t))$ is nonincreasing.

Define

$$(Tz)_j = z_j + g(z_j - z_{j-1}).$$

This operator is coordinatewise monotone. If $z \leq z'$, write

$$\alpha = z'_j - z_j, \quad \beta = z'_{j-1} - z_{j-1}.$$

If $\alpha - \beta \geq 0$, the 1-Lipschitz property gives

$$(Tz')_j - (Tz)_j \geq \alpha - (\alpha - \beta) = \beta \geq 0;$$

if $\alpha - \beta < 0$, the monotonicity of g gives the lower bound $\alpha \geq 0$.

Since $g(d_j(t))$ is nonincreasing,

$$w_j(t+1) - w_j(t) = \int_t^{t+1} g(d_j(s)) ds \leq g(d_j(t)).$$

Thus $w(t+1) \leq T(w(t))$, and monotonicity of T together with induction gives

$$W_{n,k}^P(F) = w_k(n) \leq V_{n,k}(F). \quad (7)$$

This establishes the Poisson-versus-discrete comparison without assuming the concavity properties that were omitted previously.

4 Rare-event Euler convergence

For $m \geq n$, let

$$Y_m = B_m X, \quad B_m \sim \text{Ber}(n/m).$$

Its Bellman recursion is exactly

$$v_j^{r+1} = v_j^r + \frac{n}{m} g(v_j^r - v_{j-1}^r), \quad (8)$$

the forward Euler scheme for (5) with step $h = n/m$ on the fixed interval $[0, n]$.

Because the vector field is globally 2-Lipschitz and bounded by $\mathbb{E}X$, the Euler error satisfies a discrete Grönwall bound of the form

$$\max_{r \leq m, j \leq k} |v_j^r - w_j(rn/m)| \leq \frac{n \mathbb{E}X}{2m} (e^{2n} - 1), \quad (9)$$

and hence

$$V_{m,k}(Y_m) \longrightarrow W_{n,k}^{\text{P}}(F). \quad (10)$$

For the prophet, write $q(x) = \Pr(X > x)$, so that

$$P_{m,k}(Y_m) = \int_0^\infty H_{m,k}\left(\frac{n}{m} q(x)\right) dx.$$

Pointwise,

$$H_{m,k}\left(\frac{n}{m} q\right) \longrightarrow \phi_k(nq), \quad \text{while} \quad H_{m,k}\left(\frac{n}{m} q\right) \leq nq.$$

Since $\int q = \mathbb{E}X < \infty$, dominated convergence gives

$$P_{m,k}(Y_m) \longrightarrow P_{n,k}^{\text{P}}(F) = \int_0^\infty \phi_k(nq(x)) dx. \quad (11)$$

It follows from $\gamma_{k,m} \rightarrow L_k$ that

$$W_{n,k}^{\text{P}}(F) \geq L_k P_{n,k}^{\text{P}}(F). \quad (12)$$

Thus all the finite-moment, Lipschitz, Euler, and dominated-convergence hypotheses needed in the rare-event passage are now explicit.

5 The uniform binomial–Poisson estimate

Let

$$H_{n,k}(q) = \mathbb{E}[\text{Bin}(n, q) \wedge k], \quad \phi_k(\lambda) = \mathbb{E}[\text{Pois}(\lambda) \wedge k].$$

5.1 Sign

For $p \in [0, 1]$,

$$\text{Ber}(p) \leq_{\text{cx}} \text{Pois}(p).$$

Indeed, for any convex ψ and any integer $r \geq 1$,

$$\psi(r) \geq \psi(0) + r(\psi(1) - \psi(0)).$$

Taking expectations under $\text{Pois}(p)$ yields at least

$$(1-p)\psi(0) + p\psi(1),$$

which is the Bernoulli expectation. Convex order is preserved under independent convolution, hence

$$\text{Bin}(n, p) \leq_{\text{cx}} \text{Pois}(np).$$

Since $z \mapsto z \wedge k$ is concave,

$$H_{n,k}(\lambda/n) \geq \phi_k(\lambda). \quad (13)$$

5.2 Uniform relative bound

For every fixed k there is $C_k < \infty$ such that, for all $n \geq 2k$ and $0 < \lambda \leq n$,

$$0 \leq H_{n,k}(\lambda/n) - \phi_k(\lambda) \leq \frac{C_k}{n} \phi_k(\lambda). \quad (14)$$

For $0 < \lambda \leq 1$, the elementary coupling estimate

$$d_{\text{TV}}(\text{Bin}(n, \lambda/n), \text{Pois}(\lambda)) \leq \frac{\lambda^2}{n} \quad (15)$$

follows by coupling each $\text{Ber}(p)$ with a $\text{Pois}(p)$, whose total-variation distance is

$$p(1 - e^{-p}) \leq p^2.$$

Therefore

$$H_{n,k} - \phi_k \leq \frac{k\lambda^2}{n}.$$

Since $\phi_k(\lambda) \geq 1 - e^{-\lambda} \geq \lambda/2$,

$$H_{n,k} - \phi_k \leq \frac{2k}{n} \phi_k(\lambda). \quad (16)$$

Now suppose $1 \leq \lambda \leq n/2$. For $0 \leq r < k$, write

$$p_r = e^{-\lambda} \frac{\lambda^r}{r!}, \quad b_r = \binom{n}{r} (\lambda/n)^r (1 - \lambda/n)^{n-r},$$

and set $D_r = \log(b_r/p_r)$. Then

$$D_r = \sum_{i=0}^{r-1} \log(1 - i/n) + n \left(\frac{\lambda}{n} + \log(1 - \lambda/n) \right) - r \log(1 - \lambda/n).$$

Using $n \geq 2k$ and $\lambda/n \leq 1/2$,

$$|D_r| \leq \frac{\lambda^2 + 2r\lambda + r(r-1)}{n}, \quad (D_r)_+ \leq r.$$

Hence

$$|b_r - p_r| \leq p_r e^r |D_r| \leq \frac{C_r}{n}, \quad (17)$$

where $C_r < \infty$, because an exponential times any fixed polynomial is bounded on $[1, \infty)$.

Finally, if $n/2 < \lambda \leq n$,

$$p_r \leq \frac{n^r}{r!} e^{-n/2}, \quad b_r \leq \frac{n^r}{r!} 2^{-(n-r)}.$$

Both are at most C'_r/n , uniformly in $n \geq 2k$; this includes the endpoint $\lambda = n$, where $b_r = 0$.

The identity

$$H_{n,k}(\lambda/n) - \phi_k(\lambda) = \sum_{r=0}^{k-1} (k-r)(p_r - b_r), \quad (18)$$

together with $\phi_k(\lambda) \geq 1 - e^{-1} > 1/2$ for $\lambda \geq 1$, now proves (14).

Define

$$c_{n,k} = \inf_{0 < \lambda \leq n} \frac{\phi_k(\lambda)}{H_{n,k}(\lambda/n)}.$$

Equations (13)–(14) yield

$$c_{n,k} \geq \frac{1}{1 + C_k/n}. \quad (19)$$

Combining (7), (12), and the layer-cake comparison

$$P_{n,k}^P(F) \geq c_{n,k} P_{n,k}(F)$$

gives the fully proved estimate

$$\boxed{\gamma_{k,n} \geq c_{n,k} L_k \geq \frac{L_k}{1 + C_k/n}.} \quad (20)$$

Thus the claimed $O_k(n^{-1})$ non-undershoot is repaired.

A further consequence is that L_k equals the worst marked-Poisson ratio. If

$$\rho_k = \inf_{\nu} \frac{W_k(\nu)}{P_k(\nu)},$$

the infimum taken over nonzero finite marked intensities with finite first moment, then rare-event approximation gives $L_k \leq \rho_k$, while (7), (19), and $W_k(\nu) \geq \rho_k P_k(\nu)$ give

$$\gamma_{k,n} \geq c_{n,k} \rho_k.$$

Letting $n \rightarrow \infty$ yields $L_k \geq \rho_k$, hence

$$L_k = \rho_k. \quad (21)$$

6 Exact value of $\gamma_{2,3}$

Let

$$p(t) = (9t^3 + 27t^2 + 4)^3 - 19683t^4(1+t)^2.$$

There is exactly one root $b \in (0, 1)$.

Indeed, set

$$R(t) = \frac{9t^3 + 27t^2 + 4}{t^{4/3}(1+t)^{2/3}}.$$

Then $p(t) = 0$ if and only if $R(t) = 27$, and

$$3t(1+t)(9t^3 + 27t^2 + 4) \frac{R'(t)}{R(t)} = 27t^4 + 45t^3 + 54t^2 - 24t - 16. \quad (22)$$

The polynomial on the right-hand side has exactly one positive zero by Descartes' rule of signs; it is negative at 0 and positive at 1. Hence R first decreases and then increases. Moreover,

$$\lim_{t \downarrow 0} R(t) = \infty, \quad R(1)^3 = 16000 < 27^3,$$

so the equation $R = 27$ has exactly one solution in $(0, 1)$.

Exact sign computations give

$$p(3/10) = \frac{27700956217}{10^9} > 0, \quad p(1/3) = -\frac{1016}{27} < 0,$$

so

$$\frac{3}{10} < b < \frac{1}{3}. \quad (23)$$

Define

$$g = \frac{4}{4 + 9b^3}, \quad a = \frac{27b^2}{4 + 9b^3}, \quad u = \frac{3b}{2}, \quad v = \frac{3a(1+b)}{2(1+a)}. \quad (24)$$

The bounds (23) imply

$$0 < b < u < a < v < 1. \quad (25)$$

For example, $a > u$ is equivalent to $18b > 4 + 9b^3$, while $a < 3/4$. Then

$$v > a \iff 1 + 3b > 2a, \quad v < 1 \iff a(1 + 3b) < 2,$$

and both follow from $b \in (3/10, 1/3)$ and $a < 3/4$.

Writing $D = 4 + 9b^3$, the defining equation gives the exact identities

$$\begin{aligned} ab &= 3(1 - g), & a &= 3gu^2, & ab &= 2gu^3, \\ 1 + a &= 3gv^2, & a + ab &= 2gv^3. \end{aligned} \quad (26)$$

We use independent randomized upper ranks $U_i \sim \text{Unif}(0, 1)$, with smaller ranks corresponding to larger values. Reject the first item if $U_1 > a$; otherwise reject the second if $U_2 > b$; otherwise reject the third. In the presence of atoms, the rank is generated by independent randomization within the observation's quantile interval.

The rejected-rank CDF and the selected-prefix count are

$$q_{\text{rej}}(z) = abz + a(z - b)_+ + (z - a)_+, \quad S(z) = 3z - q_{\text{rej}}(z),$$

while the prophet prefix count is

$$H(z) = 3z - z^3.$$

Using (26),

$$S(z) - gH(z) = \begin{cases} gz^3, & 0 \leq z \leq b, \\ g(z - u)^2(z + 2u), & b \leq z \leq a, \\ g(z - v)^2(z + 2v), & a \leq z \leq 1. \end{cases} \quad (27)$$

Thus $S(z) \geq gH(z)$ pointwise, and a layer-cake argument proves

$$V_{3,2}(F) \geq g P_{3,2}(F)$$

for every admissible F .

For sharpness, choose $0 < \varepsilon < b$, put

$$x = \frac{1-v}{u},$$

and define

$$Q_\varepsilon(U) = \frac{x}{\varepsilon} \mathbf{1}_{\{U \leq \varepsilon\}} + x \mathbf{1}_{\{U \leq u\}} + \mathbf{1}_{\{U \leq v\}}. \quad (28)$$

Its mean is $1+x$. If C_r denotes the minimum expected rejected value with r observations remaining, then

$$C_1 = 1+x, \quad C_2 = \mathbb{E} \min(X, C_1) = 1, \quad C_3 = \mathbb{E} \min(X, 1) = v.$$

The rank thresholds $b \in (\varepsilon, u)$ and $a \in (u, v)$ are precisely permissible tie-breaks inside the atoms at the values $1+x$ and 1 ; thus the displayed rejection policy is optimal for Q_ε .

The nested-indicator representation gives

$$P_\varepsilon = \frac{x}{\varepsilon} H(\varepsilon) + xH(u) + H(v), \quad V_\varepsilon = \frac{x}{\varepsilon} S(\varepsilon) + xS(u) + S(v).$$

Because equality holds in (27) at u and v , while $S(\varepsilon) - gH(\varepsilon) = g\varepsilon^3$,

$$V_\varepsilon = gP_\varepsilon + gx\varepsilon^2.$$

Hence

$$\frac{V_\varepsilon}{P_\varepsilon} \longrightarrow g,$$

and therefore

$$\boxed{\gamma_{2,3} = g = \frac{4}{4+9b^3}}. \quad (29)$$

7 Certified comparison with L_2

For $m \geq 5$, take

$$\Pr(X_m = 2) = \frac{1}{m}, \quad \Pr(X_m = 1) = \frac{4}{m}, \quad \Pr(X_m = 0) = 1 - \frac{5}{m}.$$

The limit is a marked Poisson process carrying value 2 at rate 1 and value 1 at rate 4.

Let $A(t)$ and $B(t)$ be its one- and two-choice values, and put $d = B - A$. Then

$$A' = (2-A)_+ + 4(1-A)_+, \quad d' = g(d) - g(A),$$

where $g(s) = (2-s)_+ + 4(1-s)_+$.

Set

$$t_0 = \frac{\log 6}{5}.$$

For $t \leq t_0$,

$$A(t) = \frac{6}{5}(1 - e^{-5t}), \quad d(t) = A(t) - 6te^{-5t}.$$

For $t = t_0 + s$, until d reaches 1,

$$A = 2 - e^{-s}, \quad d(s) = \frac{6}{5} - \frac{1}{4}e^{-s} + \left(\frac{1}{20} - t_0\right)e^{-5s}. \quad (30)$$

The first crossing $d = 1$ occurs at a unique $s_* \in (2/5, 1/2)$; after it,

$$d(s) = 2 + (s_* - e^{s_*} - s)e^{-s}.$$

Putting $S = 1 - t_0$,

$$B(1) = 4 - e^{-S}(e^{s_*} - s_* + S + 1). \quad (31)$$

The following certificate is wholly rational. For

$$S_N(y) = \sum_{j=0}^N \frac{y^j}{j!},$$

the geometric tail estimate

$$S_N(y) < e^y < S_N(y) + \frac{y^{N+1}}{(N+1)!} \frac{1}{1 - y/(N+2)} \quad (32)$$

gives

$$\begin{aligned} e^{7/4} &< 6 < e^{9/5}, & e^{13/20} &< \frac{25}{13}, \\ \frac{149}{100} &< e^{2/5} < \frac{3}{2}, & e^2 &< \frac{100}{13}, & e^{1/2} &> \frac{13}{8}, & e^{5/2} &> 12. \end{aligned} \quad (33)$$

For instance, the corresponding rational Taylor bounds are

$$\begin{aligned} e^{7/4} &< \frac{38443}{6656}, & e^{9/5} &> \frac{7542629}{1250000}, & e^{13/20} &< \frac{31997681}{16704000}, \\ \frac{559}{375} &< e^{2/5} < \frac{97}{65}, & e^2 &< \frac{23}{3}, & e^{5/2} &> \frac{110681}{9216}. \end{aligned}$$

Thus

$$\frac{7}{20} < t_0 < \frac{9}{25}.$$

Evaluating (30) and using (33) gives

$$d(2/5) < \frac{6}{5} - \frac{1}{6} - \frac{39}{1000} < 1$$

and

$$d(1/2) > \frac{6}{5} - \frac{2}{13} - \frac{31}{1200} > 1.$$

Hence $2/5 < s_* < 1/2$, as claimed. Moreover,

$$S > \frac{16}{25}, \quad e^{-S} > \frac{13}{25},$$

and $e^s - s$ is increasing for $s > 0$. Therefore

$$e^{s_*} - s_* > \frac{109}{100},$$

so (31) yields

$$B(1) < 4 - \frac{13}{25} \cdot \frac{273}{100} = \frac{6451}{2500}. \quad (34)$$

The prophet value is

$$P = \phi_2(5) + \phi_2(1) = 4 - \frac{3}{e} - 7e^{-5}.$$

Since $e > 163/60 > 27/10$,

$$\frac{3}{e} < \frac{10}{9}, \quad 7e^{-5} < \frac{700000}{14348907} < \frac{11}{225}.$$

Thus

$$P > \frac{71}{25}. \quad (35)$$

Rare-event convergence, (34), and (35) imply

$$L_2 \leq \frac{B(1)}{P} < \frac{6451}{7100}.$$

On the other hand, $b < 1/3$ gives

$$g > \frac{4}{4 + 1/3} = \frac{12}{13}, \quad \text{and} \quad \frac{6451}{7100} < \frac{12}{13}.$$

Consequently

$$\boxed{L_2 < \gamma_{2,3}}, \quad (36)$$

with no reliance on uncertified decimal ODE values.

8 Corrected tail-intensity bridge and the exact remaining gap

For $0 \leq q < 1$, define $T_{n,k}(q)$ uniquely by

$$\phi_k(T_{n,k}(q)) = H_{n,k}(q). \quad (37)$$

It satisfies

$$nq \leq T_{n,k}(q) \leq -n \log(1 - q). \quad (38)$$

The lower bound follows from (13). For the upper bound, put $\lambda = -n \log(1 - q)$, take independent $N_i \sim \text{Pois}(\lambda/n)$, and let

$$C = \sum_{i=1}^n \mathbf{1}_{\{N_i > 0\}}.$$

Then $C \sim \text{Bin}(n, q)$, while $C \leq \sum_i N_i \sim \text{Pois}(\lambda)$, proving $H_{n,k}(q) \leq \phi_k(\lambda)$.

Suppose Z is bounded with $\Pr(Z = 0) > 0$. Then

$$q(x) = \Pr(Z > x) \leq q(0) < 1,$$

so

$$\nu((x, \infty)) = T_{n,k}(q(x)) \quad (39)$$

defines an ordinary finite marked intensity. Its prophet value is exactly

$$P_k(\nu) = \int_0^\infty \phi_k(T_{n,k}(q(x))) dx = P_{n,k}(Z). \quad (40)$$

Since L_k is the worst marked-Poisson ratio,

$$W_k(\nu) \geq L_k P_{n,k}(Z). \quad (41)$$

Therefore the canonical conjecture would follow from the single inequality

$$\boxed{V_{n,k}(Z) \geq W_k(\nu)} \quad (42)$$

for every bounded $Z \geq 0$ with $\Pr(Z = 0) > 0$, where ν is defined by (39).

The earlier $q = 1$ defect is fully removable. Given an arbitrary X , put

$$a = \text{ess inf } X, \quad Y = X - a.$$

Because an optimal policy may be completed to exactly k selections,

$$P_{n,k}(X) = ka + P_{n,k}(Y), \quad V_{n,k}(X) = ka + V_{n,k}(Y). \quad (43)$$

For $\varepsilon > 0$ and $M < \infty$, let

$$Z_{\varepsilon,M} = \min\{(Y - \varepsilon)_+, M\}.$$

Since $\text{ess inf } Y = 0$, we have $\Pr(Z_{\varepsilon,M} = 0) > 0$, so (39) has finite intensity. Sending first $M \rightarrow \infty$ and then $\varepsilon \downarrow 0$ recovers Y ; the online and prophet values converge by truncation, with the ε -shift changing either value by at most $k\varepsilon$. Thus proving (42) only for these finite-rate truncations would imply the result for every admissible F ; no infinite arrival process at $q = 1$ is needed.

What remains unproved is precisely (42): equality of the prophet layer-cake curves and terminal convex-order comparisons do not control the sequential Bellman value. Hence the strongest fully proved general conclusion is

$$\boxed{\gamma_{k,n} \geq \frac{L_k}{1 + C_k/n}},$$

together with the Poisson characterization $L_k = \rho_k$ and the exact identity (29). The all- k , all- n inequality $\gamma_{k,n} \geq L_k$ remains open at the Bellman comparison (42).

References

- [1] Jiashuo Jiang, Will Ma, Jiawei Zhang. *Tightness without Counterexamples: A New Approach and New Results for Prophet Inequalities*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597734>